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MULTI-COMMAND TRANSACTIONS

Abstract

Various embodiments include methods for implementing command transactions on a two-wire bus system of a computing device, and devices for implementing the methods. Embodiments may include receiving, by a two-wire interface of a host device, an abort signal via a two-wire bus, and transmitting, by the two-wire interface of the host device, an end signal via the two-wire bus in response to receiving the abort signal. Embodiments may further include transmitting, by a two-wire interface of a bus owner device, commands via the two-wire bus prior to receiving the end signal, receiving, by the two-wire interface of the bus owner device, the end signal via the two-wire bus; and terminating, by the two-wire interface of the bus owner device, transmitting commands in response to receiving the end signal. Some embodiments may include transmitting the commands following an instance of bus arbitration and prior to a subsequent instance of bus arbitration.

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Background/Summary

BACKGROUND

[0001] Current implementations of two-wire bus systems allow communication between multiple devices of a computing device over a shared two-wire bus. A two-wire bus system implements a process of bus arbitration in which the devices having command transactions to implement via the two-wire bus attempt to gain control (or own) the two-wire bus. The bus owner device of the two-wire bus may implement one command transaction via the two-wire bus followed by another round of bus arbitration. During the subsequent bus arbitration, the previous bus owner device, if still having a command transaction to implement via the two-wire bus, can attempt to regain control of the two-wire bus. The non-bus owner devices, having a command transaction to implement via the two-wire bus, wait until the subsequent arbitration for another attempt to gain control of the two-wire bus.

SUMMARY

[0002] Various aspects provide methods include methods and apparatuses for implementing such methods for implementing command transactions on a two-wire bus system of a computing device. Various aspects may include receiving, by a two-wire interface of a host device, at least one abort signal via a two-wire bus, and transmitting, by the two-wire interface of the host device, an end signal via the two-wire bus in response to receiving the at least one abort signal.

[0003] Some aspects may further include transmitting, by a two-wire interface of a bus owner device, at least one command via the two-wire bus prior to receiving the end signal, receiving, by the two-wire interface of the bus owner device, the end signal via the two-wire bus, and terminating, by the two-wire interface of the bus owner device, transmission of commands on the two-wire bus in response to receiving the end signal. In some aspects, transmitting, by the two-wire interface of the bus owner device, the at least one command via the two-wire bus prior to receiving the end signal may include transmitting a plurality of commands, including the at least one command, following an instance of bus arbitration and prior to a subsequent instance of bus arbitration. In some aspects, the bus owner device may be a power management integrated circuit (PMIC).

[0004] In some aspects, receiving, by the two-wire interface of the host device, the at least one abort signal via the two-wire bus may include receiving the at least one abort signal from a two-wire interface of a bus owner device. In some aspects, receiving, by the two-wire interface of the host device, the at least one abort signal via the two-wire bus may include receiving the at least one abort signal from a two-wire interface of at least one non-bus owner device. In some aspects, the at least one abort signal may be a voltage transition on at least one wire of the two-wire bus.

[0005] Further aspects include a computing device including a memory and a processor and/or memory frequency device configured to perform operations of any of the methods summarized above. Further aspects include a non-transitory processor-readable storage medium having stored thereon processor-executable software instructions configured to cause a processor and/or memory frequency device to perform operations of any of the methods summarized above. Further aspects include a computing device having means for accomplishing functions of any of the methods summarized above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate example embodiments of various embodiments, and together with the

general description given above and the detailed description given below, serve to explain the features of the claims.

[0007] FIG. 1 is a component block diagram illustrating an example computing device suitable for implementing various embodiments.

[0008] FIG. 2 is a component block diagram illustrating an example two-wire bus system of the computing device suitable for implementing various embodiments.

[0009] FIGS. 3A-3C are component block and signaling diagrams illustrating examples of multi-command transactions in the two-wire bus system of the computing device suitable for implementing various embodiments.

[0010] FIGS. 4A-4C are signal and timing diagrams illustrating examples of multi-command transactions in the two-wire bus system of the computing device suitable for implementing various embodiments.

[0011] FIG. 5 is a process flow diagram illustrating an example method for implementing multi-command transactions according to an embodiment.

[0012] FIG. 6 is a component block diagram illustrating an example mobile computing device suitable for implementing various embodiments.

[0013] FIG. 7 is a component block diagram illustrating an example mobile computing device suitable for implementing various embodiments.

[0014] FIG. 8 is a component block diagram illustrating an example general purpose computing system suitable for implementing various embodiments.

DETAILED DESCRIPTION

[0015] Various embodiments will be described in detail with reference to the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. References made to particular examples and implementations are for illustrative purposes and are not intended to limit the scope of the claims.

[0016] Various embodiments include methods, and computing devices implementing such methods of implementing command transactions on a two-wire bus system of a computing device.

Embodiments may include a two-wire interface of a host device receiving at least one abort signal via a two-wire bus and transmitting an end signal via the two-wire bus in response to receiving the at least one abort signal. Some embodiments may include a two-wire interface of a bus owner device transmitting at least one command via the two-wire bus prior to receiving the end signal, receiving the end signal via the two-wire bus, and terminating transmission of commands in response to receiving the end signal. Some embodiments may include the two-wire interface of the bus owner device transmitting multiple commands via the two-wire bus following an instance of bus arbitration and prior to a subsequent instance of bus arbitration.

[0017] The term “computing device” is used herein to refer to stationary computing devices including personal computers, desktop computers, all-in-one computers, workstations, super computers, mainframe computers, embedded computers (such as in vehicles and other larger systems), computing systems within or configured for use in vehicles, servers, multimedia computers, and game consoles. The terms “computing device” and “mobile computing device” are used interchangeably herein to refer to any one or all of cellular telephones, smartphones, personal or mobile multi-media players, personal data assistants (PDA's), laptop computers, tablet computers, convertible laptops/tablets (2-in-1 computers), smartbooks, ultrabooks, netbooks, palm-top computers, wireless electronic mail receivers, multimedia Internet enabled cellular telephones, mobile gaming consoles, wireless gaming controllers, and similar personal electronic devices that include a memory, and a programmable processor.

[0018] Various embodiments are described in terms of code, e.g., processor-executable instructions, for ease and clarity of explanation, but may be similarly applicable to any data, e.g., code, program data, or other information stored in memory. The terms “code,” “data,” and “information” are used interchangeably herein and are not intended to limit the scope of the claims and descriptions to the

types of code, data, or information used as examples in describing various embodiments.

[0019] Current implementations of two-wire bus systems allow communication between multiple devices of a computing device over a shared two-wire bus. A two-wire bus system implements a process of bus arbitration in which the devices having command transactions to implement via the two-wire bus attempt to gain control (or own) the two-wire bus to permit transmitting commands or data. The bus owner device of the two-wire bus may implement one command transaction via the two-wire bus followed by another round of bus arbitration. During the subsequent bus arbitration, the previous bus owner device, if still having a command transaction to implement via the two-wire bus, can attempt to regain control of the two-wire bus. The non-bus owner devices, having a command transaction to implement via the two-wire bus, must wait until the subsequent bus arbitration for another attempt to gain control of the two-wire bus.

[0020] Each instance of bus arbitration before each command transaction at the two-wire bus system incur non-negligible time costs. Given that the two-wire bus system can implement only one command transaction at a time, the time cost for each bus arbitration before each command transaction is cumulative over time. The cumulative time cost for one or more processes of the devices connected to the two-wire bus affects the time performance of the processes and the performance perception of a user of the computing device. For example, a process having multiple command transactions incurs the time cost for each bus arbitration for each command transaction, even without interruption for other processes to use the two-wire bus. With interruption for other processes to use the two-wire bus the time cost for the process increases over the time cost without interruption for other processes to use the two-wire bus.

[0021] Embodiments of multi-command transactions overcome the foregoing problems by enabling transmission of multiple serial commands by a bus owner device between instances of bus arbitration. A two-wire bus system may be configured to implement multi-command transactions. Two-wire bus interfaces of devices connected to a two-wire bus may be configured to transmit and/or receive abort signals and/or end signals. The abort signal may be configured to trigger transmission of the end signal. The end signal may be configured to trigger termination of transmitting commands by the bus owner device. The two-wire bus interface of the bus owner device may be configured to transmit multiple serial commands until receipt of the end signal rather than transmitting only one command before vying for control of the two-wire bus again in bus arbitration for transmitting a subsequent command.

[0022] The two-wire bus system may include multiple devices connected to the two-wire bus. Each of the devices may have a two-wire bus interface configured for communicating via the two-wire bus. The two-wire bus interfaces of the devices may be configured to implement aspects of multi-command transactions. The two-wire bus interfaces of the devices may be configured to transmit multiple serial commands and to transmit abort signals. The two-wire bus interface of a host device may be configured to receive the abort signals and to transmit the end signal in response to receiving an abort signal. The two-wire bus interfaces of the devices may be configured to receive the end signal and terminate transmitting the multiple serial commands in response.

[0023] Through bus arbitration by the two-wire bus system, one of the devices may gain control over the two-wire bus, becoming the “bus owner” device with permission to transmit commands over the two-wire bus. Some devices may not vie for control of the two-wire bus during arbitration. Other devices may not win control of the two-wire bus during arbitration. Those devices that do not achieve control of the two-wire bus are “non-bus owners” that do not have permission to transmit commands over the two-wire bus. The bus owner device may transmit at least one command, including multiple serial commands, over the two-wire bus. Any of the devices, the bus owner device and/or the non-bus owner devices, may transmit an abort signal over the two-wire bus. The host device may receive the abort signal and respond by transmitting the end signal over the two-wire bus. The bus owner device may receive the end signal and respond by terminating transmission of commands. Following the bus owner device terminating transmission of

commands, the two-wire bus system may implement a subsequent bus arbitration.

[0024] In a non-limiting example, during a boot process of a power management integrated circuitry (PMIC), PMIC random access memory (RAM) data may be loaded from a system-on-chip (SoC) to the PMIC via a PMIC interface (e.g., system power management interface (SPMI)). Implementing multi-command transactions during the boot process of the PMIC can reduce the time of the boot process as compared to the boot process implementing single command transactions with bus arbitration implemented for each command transaction. For example, an amount of data of $70,000 \times$ bytes of data results in approximately 8500×8 -byte SPMI extended register write long commands that may take approximately 41 ms without multi-command transactions. Implementing multi-command transactions, eliminating the repeated bus arbitration, the time to transmit the $70,000 \times$ bytes of data may reduce to approximately 37.8 ms, which is an approximately 8% latency reduction.

[0025] FIG. 1 illustrates a system including a computing device **10** suitable for use with various embodiments. With reference to FIG. 1, the computing device **10** may include a system-on-chip (SoC) **12** with a processor **14**, a memory **16**, a communication interface **18**, a storage memory interface **20**, a memory interface **34**, a power manager **28**, a clock controller **30**, a peripheral device interface **38**, and an interconnect **32**. The computing device **10** may further include a communication component **22**, such as a wired or wireless modem, a storage memory **24**, an antenna **26** for establishing a wireless communication link, a memory **36**, and a peripheral device **40**. The processor **14** may include any of a variety of processing devices, for example a number of processor cores.

[0026] The term “system-on-chip” (SoC) is used herein to refer to a set of interconnected electronic circuits typically, but not exclusively, including a processing device, a memory, and a communication interface. A processing device may include a variety of different types of processors **14** and processor cores, such as a general purpose processor, a central processing unit (CPU), a digital signal processor (DSP), a graphics processing unit (GPU), an accelerated processing unit (APU), a secure processing unit (SPU), a subsystem processor of specific components of the computing device, such as an image processor for a camera subsystem or a display processor for a display, an auxiliary processor, a single-core processor, a multicore processor, a controller, and a microcontroller. A processing device may further embody other hardware and hardware combinations, such as a field programmable gate array (FPGA), an application-specific integrated circuit (ASIC), other programmable logic device, discrete gate logic, transistor logic, performance monitoring hardware, watchdog hardware, and time references. Integrated circuits may be configured such that the components of the integrated circuit reside on a single piece of semiconductor material, such as silicon.

[0027] An SoC **12** may include one or more processors **14**. The computing device **10** may include more than one SoC **12**, thereby increasing the number of processors **14** and processor cores. The computing device **10** may also include processors **14** that are not associated with an SoC **12**. The processors **14** may each be configured for specific purposes that may be the same as or different from other processors **14** of the computing device **10**. One or more of the processors **14** and processor cores of the same or different configurations may be grouped together. A group of processors **14** or processor cores may be referred to as a multi-processor cluster.

[0028] The memory **16**, **36** for the SoC **12** may be a volatile or non-volatile memory configured for storing data and processor-executable code for access by the processor **14**. The computing device **10** and/or SoC **12** may include one or more memories **16**, **36** configured for various purposes. One or more memories **16**, **36** may include volatile memories such as random access memory (RAM) or main memory, including static RAM (SRAM) and/or dynamic RAM (DRAM), or cache memory. These memories **16**, **36** may be configured to temporarily hold a limited amount of data received from a data sensor or subsystem, data and/or processor-executable code instructions that are requested from a non-volatile memory **16**, **24**, loaded to the memories **16**, **36** from the non-volatile

memory **16, 24** in anticipation of future access based on a variety of factors, and/or intermediary processing data and/or processor-executable code instructions produced by the processor **14** and temporarily stored for future quick access without being stored in non-volatile memory **16, 24**. The memory **16, 36** may include multiple physical memory components, such as memory chips, that may be logically combined and/or separated to form the memory **16, 36**. The memory interface **34** and the memory **36** may work in unison to allow the computing device **10** to load and retrieve data and processor-executable code on the memory **36**.

[0029] The storage memory interface **20** and the storage memory **24** may work in unison to allow the computing device **10** to store data and processor-executable code on a non-volatile storage medium. The storage memory **24** may be configured much like an embodiment of the memory **16** in which the storage memory **24** may store the data or processor-executable code for access by one or more of the processors **14**. The storage memory **24**, being non-volatile, may retain the information after the power of the computing device **10** has been shut off. When the power is turned back on and the computing device **10** reboots, the information stored on the storage memory **24** may be available to the computing device **10**. The storage memory **24** may include multiple physical memory components, such as storage memory drives, chips, discs, etc., that may be logically combined and/or separated to form the storage memory **24**. The storage memory interface **20** may control access to the storage memory **24** and allow the processor **14** to read data from and write data to the storage memory **24**.

[0030] The power manager **28** may be configured to control power states of one or more power rails (not shown) for power delivery to the components of the SoC **12**. In some embodiments, the power manager **28** may be configured to control amounts of power provided to the components of the SoC **12**. For example, the power manager **28** may be configured to control connections between components of the SoC **12** and the power rails. As another example, the power manager **28** may be configured to control amounts of power on the power rails connected to the components of the SoC **12**. The power manager **28** may be configured as a power management integrated circuit (power management ICs or PMIC).

[0031] A clock controller **30** may be configured to control clock signals transmitted to the components of the SoC **12**. For example, the clock controller **30** may gate a component of the SoC **12** by disconnecting the component of the SoC **12** from a clock signal and may ungate the component of the SoC **12** by connecting the component of the SoC **12** to the clock signal.

[0032] A peripheral device interface **38** may enable components of the SoC **12**, such as the processor **14** and/or the memory **16**, to communicate with a peripheral device **40**. The peripheral device interface **38** may provide and manage physical and logical connections between the components of the SoC **12** and the peripheral device **40**. The peripheral device interface **38** may also manage communication between the components of the SoC **12** and the peripheral device **40**, such as by directing and/or allowing communications between transmitter and receiver pairs of the components of the SoC **12** and the peripheral device **40** for a communication. The communications may include transmission of memory access commands, addresses, data, interrupt signals, state signals, etc. A peripheral device **40** may be any component of the computing device **10** separate from the SoC **12**, such as a processor, a memory, a subsystem, etc. In some embodiments, the peripheral device interface **38** may include a PCIe root complex and may enable PCIe protocol communication between the components of the SoC **12** and the peripheral device **40**.

[0033] The interconnect **32** may be a communication fabric, such as a communication bus, configured to communicatively connect the components of the SoC **12**. The interconnect **32** may transmit signals between the components of the SoC **12**. In some embodiments, the interconnect **32** may be configured to control signals between the components of the SoC **12** by controlling timing and/or transmission paths of the signals.

[0034] Some or all of the components, including components of the SoC **12**, connected to the SoC **12**, and the SoC **12**, of the computing device **10** may be arranged differently, separated, and/or

combined while still serving the functions of the various embodiments. The computing device **10** may not be limited to one of each of the components, and multiple instances of each component may be included in various configurations of the computing device.

[0035] FIG. 2 illustrates an example two-wire bus system **200** of the computing device (e.g., computing device **10**) suitable for implementing various embodiments. With reference to FIGS. 1 and 2, components of the computing device may include components of the two-wire bus system **200**, including a two-wire bus **202** (e.g., interconnect **32**) connecting multiple devices **208**, **210**, **212a**, **212b** (e.g., SoC **12**, processor **14**, memory **16**, communication interface **18**, storage memory interface **20**, communication component **22**, storage memory **24**, power manager **28**, clock controller **30**, memory interface **34**, memory **36**, peripheral device interface **38**, peripheral device **40**). The two-wire bus **202** may include a clock wire **204** and a data wire **206**. The clock wire may be configured to transmit a clock signal and the data wire **206** may be configured to transmit data signals.

[0036] The multiple devices **208**, **210**, **212a**, **212b** may each have a two-wire interface **214**, including a processor system of one or more processors (e.g., processor **14**), configured to receive signals transmitted by and/or transmit signals over the two-wire bus **202** per a two-wire bus protocol. For example, the two-wire bus protocol may include I2C, SPI, UART, SPMI, etc.

[0037] While descriptions of the example of FIG. 2 may be expressed in terms of the multiple devices **208**, **210**, **212a**, **212b**, it may be understood that the functions of the multiple devices **208**, **210**, **212a**, **212b** of the example may be implemented by the two-wire interfaces **214** of the corresponding devices **208**, **210**, **212a**, **212b**.

[0038] The multiple devices **208**, **210**, **212a**, **212b** may include a host device (e.g., processor **12**). Any of the multiple devices **208**, **210**, **212a**, **212b** may, at various times, be a bus owner device **210** and/or a non-bus owner device **212a**, **212b**. The bus owner device **210** is a device that controls, or “owns,” the two-wire bus **202** as a result of a bus arbitration process. The non-bus owner devices **212a**, **212b** are devices that do not control, or do not “own,” the two-wire bus **202** as a result of a bus arbitration process, by either not vying for ownership or failing to achieve ownership of the two-wire bus **202**. The bus owner device **210** may transmit commands over the data wire **206** of the two-wire bus **202**. The non-bus owner devices **212a**, **212b** may not transmit commands over the data wire **206** of the two-wire bus **202**.

[0039] The two-wire interface **214** of each of the multiple devices **208**, **210**, **212a**, **212b** may be configured to implement multi-command transactions. The two-wire interface **214** of the bus owner device **210** may enable the bus owner device **210** to send at least one command, including multiple commands serially, over the data wire **206** of the two-wire bus **202** between instances of bus arbitration.

[0040] The two-wire interface **214** of any of the multiple devices **208**, **210**, **212a**, **212b** may enable any of the multiple devices **208**, **210**, **212a**, **212b** to generate and transmit an abort signal over the data wire **206** of the two-wire bus **202**. The abort signal may be a voltage transition on at least one wire of the two-wire bus. The abort signal may be triggered in response to various events. For example, the bus owner device **210** may generate and transmit an abort signal in response to completing transmitting all of and/or a last of commands the bus owner device **210** has queued. As another example, the non-bus owner device **212a**, **212b**, which may include the host device **208**, may generate and transmit an abort signal in response to having queued a command with a priority level for which the non-bus owner device **212a**, **212b** is configured to interrupt transmitting one or more commands by the bus owner device **210**. The abort signal may be configured to trigger the host device **208** to generate and transmit the end signal.

[0041] The two-wire interface **214** of the host device **208** may enable the host device **208** to generate and transmit an end signal over the data wire **206** of the two-wire bus **202**. In response to receiving the abort signal from one or more of the multiple devices **208**, **210**, **212a**, **212b** over the data wire **206** of the two-wire bus **202**, the host device **208** may generate and transmit an end

signal. The end signal may be configured to trigger the bus owner device **210** to terminate transmitting commands. The end signal may also be configured to trigger the two-wire bus system **200** to implement bus arbitration to give one or more of the non-bus owner devices **212a**, **212b** to gain ownership of the two-wire bus **202**.

[0042] The two-wire interface **214** of the bus owner device **210** may enable the bus owner device **210** to terminate transmitting commands over the data wire **206** of the two-wire bus **202**. The bus owner device **210** may receive the end signal and, in response, terminate transmitting any commands that the bus owner device **210** may have queued.

[0043] FIGS. 3A-3C illustrate examples of multi-command transactions in the two-wire bus system (e.g., two-wire bus system **200** in FIG. 2) of the computing device (e.g., computing device **10**) suitable for implementing various embodiments. With reference to FIGS. 1-3C, multiple devices **308**, **310**, **312**, (e.g., SoC **12**, processor **14**, memory **16**, communication interface **18**, storage memory interface **20**, communication component **22**, storage memory **24**, power manager **28**, clock controller **30**, memory interface **34**, memory **36**, peripheral device interface **38**, peripheral device **40**, host device **208**, bus owner device **210**, non-bus owner device **212a**, **212b** in FIG. 2) may be configured to implement multi-command transactions over a two-wire bus (e.g., interconnect **32**, two-wire bus **202** in FIG. 2). The multiple devices **308**, **310**, **312** may each have a two-wire interface **314** (e.g., two-wire interface **214** in FIG. 2) to the two wire bus, including a processor system of one or more processors (e.g., processor **14**), configured to enable the multiple devices **308**, **310**, **312** to implement multi-command transactions.

[0044] While descriptions of the examples of FIGS. 3A-3C may be expressed in terms of the multiple devices **308**, **310**, **312**, the processes of the examples may be implemented by the two-wire interfaces **314** of the corresponding devices **308**, **310**, **312**.

[0045] FIG. 3A illustrates, in process **320**, an example instance of bus arbitration on the two-wire bus system in which the bus owner device **310** gains control of the two-wire bus. The bus owner device **310** may transmit at least one command over the two-wire bus in process **322**. While the bus owner device **310** has not received an end signal and has at least one command queued for transmission, the bus owner device **310** may continue to serially transmit commands over the two-wire bus.

[0046] While the bus owner device **310** continues to serially transmit commands, at least one of multiple non-bus owner devices **312** may generate and transmit an abort signal over the two-wire bus in process **324a**. Generating and transmitting the abort signal by the non-bus owner device **312** may be triggered by the non-bus owner device **312** having queued a command to transmit that has a priority for which the non-bus owner device **312** may interrupt the bus-owner device **310** transmitting commands over the two-wire bus.

[0047] Also, while the bus owner device **310** continues to serially transmit commands, the host device **308** may receive the abort signal and, in response, generate and transmit an end signal over the two-wire bus in process **326**. The bus owner device **310** may receive the end signal and, in response, terminate sending any commands that the bus owner device **310** may have queued for transmission in process **328**. As a result of the end signal and termination of transmitting commands by the bus owner device **310**, a subsequent instance of bus arbitration on the two-wire bus may be implemented in process **320**. The bus owner device **310** may lose and/or relinquish control over and/or ownership of the two-wire bus during bus arbitration. The bus owner device **310** may regain control over and/or ownership of the two-wire bus and/or the non-bus owner device **312** may gain control over and/or ownership of the two-wire bus as a result of the bus arbitration.

[0048] FIG. 3B illustrates, in process **320**, an example instance of bus arbitration in the two-wire bus system in which the bus owner device **310** gains control of the two-wire bus. The bus owner device **310** may transmit at least one command over the two-wire bus in process **322**. While the bus owner device **310** has not received an end signal and has at least one command queued for

transmission, the bus owner device **310** may continue to serially transmit commands over the two-wire bus.

[0049] The bus owner device **310** may generate and transmit an abort signal over the two-wire bus in process **324b**. In some embodiments, the bus owner device **310** may generate the abort signal as a voltage transition on at least one wire of the two-wire bus. Generating and transmitting the abort signal by the bus owner device **310** may be triggered by the bus owner device **310** not having queued a command to transmit.

[0050] The host device **308** may receive the abort signal and, in response, generate and transmit an end signal over the two-wire bus in process **326**. The bus owner device **310** may receive the end signal and, in response, terminate sending any commands that the bus owner device **310** may have queued for transmission in process **328**. As a result of the end signal and termination of transmitting commands by the bus owner device **310**, a subsequent instance of bus arbitration may occur in the two-wire bus system in process **320**. The bus owner device **310** may lose and/or relinquish control over and/or ownership of the two-wire bus during bus arbitration. The bus owner device **310** may regain control over and/or ownership of the two-wire bus and/or the non-bus owner device **312** may gain control over and/or ownership of the two-wire bus as a result of the bus arbitration.

[0051] FIG. **3C** illustrates, in process **320**, an example instance of bus arbitration in the two-wire bus system in which the bus owner device **310** gains control of the two-wire bus. The bus owner device **310** may transmit at least one command over the two-wire bus in process **322**. While the bus owner device **310** has not received an end signal and has at least one command queued for transmission, the bus owner device **310** may continue to serially transmit commands over the two-wire bus.

[0052] While the bus owner device **310** continues to serially transmit commands, the host device owner device **308** may generate and transmit an abort signal to the host device owner device **308** in process **324c**. Generating and transmitting the abort signal by the host device **308** may be triggered by the host device **308** having queued a command to transmit that has a priority for which the host bus owner device **308** may interrupt the bus-owner device **310** transmitting commands over the two-wire bus.

[0053] Also, while the bus owner device **310** continues to serially transmit commands, the host device **308** may receive the abort signal and, in response, generate and transmit an end signal over the two-wire bus in process **326**. The bus owner device **310** may receive the end signal and, in response, terminate sending any commands that the bus owner device **310** may have queued for transmission in process **328**. As a result of the end signal and termination of transmitting commands by the bus owner device **310**, a subsequent instance of bus arbitration may occur in the two-wire bus system in process **320**. The bus owner device **310** may lose and/or relinquish control over and/or ownership of the two-wire bus during bus arbitration. The bus owner device **310** may regain control over and/or ownership of the two-wire bus and/or the non-bus owner device **312** may gain control over and/or ownership of the two-wire bus as a result of the bus arbitration.

[0054] FIGS. **4A-4C** illustrate examples of multi-command transactions in a two-wire bus system (e.g., two-wire bus system **200** in FIG. **2**) of the computing device (e.g., computing device **10**) suitable for implementing various embodiments. With reference to FIGS. **1-4C**, multiple devices (e.g., SoC **12**, processor **14**, memory **16**, communication interface **18**, storage memory interface **20**, communication component **22**, storage memory **24**, power manager **28**, clock controller **30**, memory interface **34**, memory **36**, peripheral device interface **38**, peripheral device **40**, host device **208**, **308**, bus owner device **210**, **310**, non-bus owner device **212a**, **212b**, **312**) may be configured to implement multi-command transactions over a two-wire bus (e.g., interconnect **32**, two-wire bus **202**, **302**). The multiple devices may each have a two-wire interface (e.g., two-wire interface **214**, **314**), including a processor system of one or more processors (e.g., processor **14**), configured to enable the multiple devices to implement multi-command transactions.

[0055] Signals may be transmitted via the clock wire (e.g., clock wire **204** in FIG. 2) and data wire (data wire **206**) of the two-wire bus by the multiple devices. The clock wire may transmit a clock signal **402a**, **402b**, **402c**, **402d**, **402e**. The data wire may transmit data signals **404a**, **404b**, **404c**, **404d**, **404e**.

[0056] The examples of FIGS. 4A-4C may be expressed in terms of the multiple devices, it may be understood that the signaling of the examples may be implemented by the two-wire interfaces of the corresponding devices.

[0057] The example in FIG. 4A illustrates a command transaction over the two-wire bus as implemented by the bus owner device (e.g., SoC **12**, processor **14**, memory **16**, communication interface **18**, storage memory interface **20**, communication component **22**, storage memory **24**, power manager **28**, clock controller **30**, memory interface **34**, memory **36**, peripheral device interface **38**, peripheral device **40**, bus owner device **210**, **310**). Transmitting a command during a command transaction via the data signals **404a** may include transmitting a destination identity **406** and a command **408**. The destination identity **406** may identify a targeted non-bus owner device (e.g., SoC **12**, processor **14**, memory **16**, communication interface **18**, storage memory interface **20**, communication component **22**, storage memory **24**, power manager **28**, clock controller **30**, memory interface **34**, memory **36**, peripheral device interface **38**, peripheral device **40**, non-bus owner device **212a**, **212b**, **312**) at which to implement the command **408**. The command **408** may include the instructions for interpretation by the targeted non-bus owner device for implementing the command **408**. In some embodiments, the command **408** may include a parity value configured for error checking. The destination identity **406** and command **408** may be transmitted by the bus owner device and received by the non-bus owner device.

[0058] Following transmitting the command **408**, data **410** for the command may be transmitted via the data signals **404b**. The command **408** may be for storing and/or operating on the data **410**. In some embodiments, the data **410** may include a parity value configured for error checking. The data **410** may be transmitted by the bus owner device and received by the non-bus owner device.

[0059] Following transmitting the data **410**, error checking **412** may be implemented by the non-bus owner device. The bus owner device may transmit error checking data and/or commands **412** via the data signals **404c** for the non-bus owner device to implement error checking for the command **408**.

[0060] The example in FIG. 4B illustrates abort signal checking when an abort signal is absent. The bus owner device (e.g., SoC **12**, processor **14**, memory **16**, communication interface **18**, storage memory interface **20**, communication component **22**, storage memory **24**, power manager **28**, clock controller **30**, memory interface **34**, memory **36**, peripheral device interface **38**, peripheral device **40**, bus owner device **210**, **310**) may implement bus parking **420**, driving the data signals **404d** low. The non-bus owner device (e.g., SoC **12**, processor **14**, memory **16**, communication interface **18**, storage memory interface **20**, communication component **22**, storage memory **24**, power manager **28**, clock controller **30**, memory interface **34**, memory **36**, peripheral device interface **38**, peripheral device **40**, non-bus owner device **212a**, **212b**, **312**) may transmit an acknowledgement **422** of the command transaction via the data signals **404d**. The bus owner device may drive the data signals **404d** low **424** again. The host device (e.g., processor **12**, host device **208**, **308**) may check for an abort signal at the end of or following each command transaction, which in the example of FIG. 4B is not transmitted via the data signals **404d**.

[0061] In some embodiments, the bus owner device may check for the abort signal at the end of or following each command transaction and before implementing a subsequent command transaction, in a repetition of the command transaction illustrated in FIG. 4A. In some embodiments, the bus owner device may be configured to allow time for any of the multiple devices to transmit an abort signal and for the host device to send an end signal before implementing a subsequent command transaction.

[0062] A multi-command transaction may include repeating the command transaction illustrated in

FIG. 4A and the abort signal check in FIG. 4B in the absence of an abort signal.

[0063] The example in FIG. 4C illustrates abort signal checking when an abort signal is present. The bus owner device (e.g., SoC 12, processor 14, memory 16, communication interface 18, storage memory interface 20, communication component 22, storage memory 24, power manager 28, clock controller 30, memory interface 34, memory 36, peripheral device interface 38, peripheral device 40, bus owner device 210, 310) may implement bus parking 420, driving the data signals 404e low. The non-bus owner device (e.g., SoC 12, processor 14, memory 16, communication interface 18, storage memory interface 20, communication component 22, storage memory 24, power manager 28, clock controller 30, memory interface 34, memory 36, peripheral device interface 38, peripheral device 40, non-bus owner device 212a, 212b, 312) may transmit an acknowledgement 422 of the command transaction via the data signals 404e. The bus owner device may drive the data signals 404e low 424 again.

[0064] An abort signal 440 may be transmitted by any of the multiple devices via the data signals 404e. The host device (e.g., processor 12, host device 208, 308) may check for the abort signal 440 at the end of or following each command transaction. The host device may check for the abort signal before a subsequent command transaction, in a repetition of the command transaction illustrated in FIG. 4A. The host device may receive the abort signal 440. In some embodiments, the bus owner device may check for the abort signal at the end of or following each command transaction and before implementing a subsequent command transaction. The bus owner device may receive the abort signal 440 and terminate the multi-command transaction, preventing any subsequent command transactions. In some embodiments, the bus owner device may be configured to allow for time for any of the multiple devices to transmit an abort signal and for the host device to send an end signal before implementing a subsequent command transaction. The bus owner device may drive the data signals 404e low 442 again.

[0065] The host device may implement signaling for a clock handover 444 via the data signals 404e. The host device may also transmit an end signal 446 via the data signals 404e and that may be received by the bus owner device via the data signals 404e. Following transmitting the end signal 446, the two-wire bus may idle 448.

[0066] Before implementing another command transaction, as in the example illustrated in FIG. 4A, the two-wire bus system may implement bus arbitration following the abort signal checking when the abort signal is present.

[0067] FIG. 5 illustrates an example method for implementing multi-command transactions according to an embodiment. With reference to FIGS. 1-5, the method 500 may be implemented in a computing device (e.g., computing device 10), in hardware (e.g., processor 12, host device 208, 308, bus owner device 210, 310, two-wire interface 214, 314), in software executing in a processor (e.g., processor 12, host device 208, 308, bus owner device 210, 310, two-wire interface 214, 314), or in a combination of a software-configured processor and dedicated hardware, such as the two-wire bus system (e.g., two-wire bus system 200), that includes other individual components, such as various memories/caches. In order to encompass the alternative configurations enabled in various embodiments, the hardware implementing the method 500 is referred to herein as a “multi-command transaction device.”

[0068] In block 502, the multi-command transaction device may transmit a command (e.g., destination identity 406, command 408, data 410, error checking data and/or commands 412 in FIG. 4A). The command may be sent by multi-command transaction device that owns the two-wire bus (e.g., two-wire bus 202, 302) over the two-wire bus. In some embodiments, the multi-command transaction device transmitting the command in block 502 may be a bus owner device (e.g., bus owner device 210, 310) and/or two-wire interface (e.g., two-wire interface 214, 314), such as a two-wire interface of the bus owner device.

[0069] In optional block 504, the multi-command transaction device may issue an abort signal (e.g., abort signal 440). The abort signal may be issued by any configuration of the multi-command

transaction device. In some embodiments, the abort signal may be issued internally with the multi-command transaction device. In some embodiments, the abort signal may be transmitted via the two-wire bus. In some embodiments, the abort signal may be issued as a voltage transition on at least one wire of the two-wire bus. For example, the multi-command transaction device that does not own the two-wire bus may issue an abort signal in response to having queued a command with a priority level for which the multi-command transaction device is configured to interrupt transmitting one or more commands by the multi-command transaction device that owns the two-wire bus. As another example, the multi-command transaction device that owns the two-wire bus may issue an abort signal in response to not having queued a command to transmit. Issuing an abort signal in block **504** is optional because issuance of the abort signal is contingent on factors that may or may not occur in various instances. In some embodiments, the multi-command transaction device issuing an abort signal in optional block **504** may be a host device (e.g., processor **12**, host device **208**, **308**), the bus owner device, a non-bus owner device (e.g., non-bus owner device **212a**, **212b**, **312**) and/or the two-wire interface, such as a two-wire interface of the host device, the bus owner device, and/or the non-bus owner device.

[0070] In optional block **506**, the multi-command transaction device may receive the abort signal. The abort signal may be received via the two-wire bus. In some embodiments, the abort signal may be a voltage transition on at least one wire of the two-wire bus. Receiving the abort signal in block **506** is optional because issuance of the abort signal in block **504** is contingent on factors that may or may not occur in various instances. In some embodiments, the multi-command transaction device receiving the abort signal in optional block **506** may be the host device and/or the two-wire interface, such as a two-wire interface of the host device.

[0071] In determination block **508**, the multi-command transaction device may determine whether the abort signal is received. There may be instances in which no abort signal was issued and therefore may not be received. There may be instances in which the abort signal was issued and therefore may be received. In some embodiments, the multi-command transaction device determining whether the abort signal is received in determination block **508** may be the host device and/or the two-wire interface, such as a two-wire interface of the host device.

[0072] In response to determining that the abort signal is not received (i.e., determination block **508**="No"), the multi-command transaction device may transmit a command in block **502**. The multi-command transaction device may transmit another command sequentially to a previous command transmitted in a previous implementation of block **502** between which no abort signal or end signal is issued, or bus arbitration is implemented between transmitting of commands. In some embodiments, the multi-command transaction device transmitting the command in block **502** may be the bus owner device and/or the two-wire interface, such as a two-wire interface of the bus owner device. Thus, until an abort signal is received, the multi-command transaction device may transmit a command over the two wire bus.

[0073] In response to determining that the abort signal is received (i.e., determination block **508**="Yes"), the multi-command transaction device may transmit an end signal (e.g., end signal **446**) in block **510**. The abort signal may be configured to trigger the multi-command transaction device to generate and transmit the end signal. The end signal may be transmitted via the two-wire bus. In some embodiments, the multi-command transaction device transmitting the end signal in block **510** may be the host device and/or the two-wire interface, such as a two-wire interface of the host device.

[0074] In block **512**, the multi-command transaction device may receive the end signal. The end signal may be received via the two-wire bus. In some embodiments, the multi-command transaction device receiving the end signal in block **512** may be the bus owner device and/or the two-wire interface, such as a two-wire interface of the bus owner device.

[0075] In block **514**, the multi-command transaction device may terminate transmitting commands. The end signal may be configured to trigger the multi-command transaction device to terminate

transmitting commands. Regardless of whether the multi-command transaction device may have further commands in queue to transmit, the multi-command transaction device may end transmitting the commands in queue until the multi-command transaction device owns the two-wire bus again. In some embodiments, the multi-command transaction device terminating transmission of commands in block **514** may be the bus owner device and/or the two-wire interface, such as a two-wire interface of the bus owner device.

[0076] To resume transmitting commands, the multi-command transaction device may have to win a bus arbitration to regain control of the two-wire bus. Winning a bus arbitration may enable the multi-command transaction device to restart the method **500**, transmitting a command in block **502**.

[0077] A system in accordance with the various embodiments (including, but not limited to, embodiments described above with reference to FIGS. **1-5**) may be implemented in a wide variety of computing systems including mobile computing devices, an example of which suitable for use with the various embodiments is illustrated in FIG. **6**. The mobile computing device **600** may include a processor **602** coupled to a touchscreen controller **604** and an internal memory **606**. The processor **602** may be one or more multicore integrated circuits designated for general or specific processing tasks. The internal memory **606** may be volatile or non-volatile memory, and may also be secure and/or encrypted memory, unsecure and/or unencrypted memory, or any combination thereof. Examples of memory types that can be leveraged include but are not limited to DDR, Low-Power DDR (LPDDR), Graphics DDR (GDDR), WIDEIO, RAM, Static RAM (SRAM), Dynamic RAM (DRAM), Parameter RAM (P-RAM), Resistive RAM (R-RAM), Magnetoresistive RAM (M-RAM), Spin-Transfer Torque RAM (STT-RAM), and embedded DRAM.

[0078] The touchscreen controller **604** and the processor **602** may also be coupled to a touchscreen panel **612**, such as a resistive-sensing touchscreen, capacitive-sensing touchscreen, infrared sensing touchscreen, etc. Additionally, the display of the mobile computing device **600** need not have touch screen capability.

[0079] The mobile computing device **600** may have one or more radio signal transceivers **608** (e.g., Peanut, Bluetooth, ZigBee, Wi-Fi, RF radio) and antennae **610**, for sending and receiving communications, coupled to each other and/or to the processor **602**. The processor **602** may also be coupled to a cellular network wireless modem **609** that enables communication via a cellular network (e.g., a 5G network) via the antenna **610**. The transceivers **608** and antennae **610** may be used with the above-mentioned circuitry to implement the various wireless transmission protocol stacks and interfaces.

[0080] The mobile computing device **600** may also include speakers **614** for providing audio outputs. The mobile computing device **600** may also include a housing **620**, constructed of a plastic, metal, or a combination of materials, for containing all or some of the components described herein. The mobile computing device **600** may include a power source **622** coupled to the processor **602**, such as a disposable or rechargeable battery. The rechargeable battery may also be coupled to the peripheral device connection port to receive a charging current from a source external to the mobile computing device **600**. The mobile computing device **600** may also include a physical button **624** for receiving user inputs. The mobile computing device **600** may also include a power button **626** for turning the mobile computing device **600** on and off.

[0081] A system in accordance with the various embodiments (including, but not limited to, embodiments described above with reference to FIGS. **1-5**) may be implemented in a wide variety of computing systems including a laptop computer **700**, an example of which is illustrated in FIG. **7**. Many laptop computers include a touchpad touch surface **717** that serves as the computer's pointing device, and thus may receive drag, scroll, and flick gestures similar to those implemented on computing devices equipped with a touch screen display and described above. A laptop computer **700** will typically include a processor **702** coupled to volatile memory **712** and a large capacity nonvolatile memory, such as a disk drive **713** of Flash memory. Additionally, the computer **700** may have one or more antenna **708** for sending and receiving electromagnetic radiation that

may be connected to a wireless data link and/or cellular telephone transceiver **716** coupled to the processor **702**. The computer **700** may also include a floppy disc drive **714** and a compact disc (CD) drive **715** coupled to the processor **702**. In a notebook configuration, the computer housing includes the touchpad **717**, the keyboard **718**, and the display **719** all coupled to the processor **702**. Other configurations of the computing device may include a computer mouse or trackball coupled to the processor (e.g., via a USB input) as are well known, which may also be used in conjunction with the various embodiments.

[0082] A system in accordance with the various embodiments (including, but not limited to, embodiments described above with reference to FIGS. **1-5**) may also be implemented in fixed computing systems, such as any of a variety of generalized or specialize computing systems, including computing systems that may be implemented in vehicles. An example computing system **800** is illustrated in FIG. **8**. Such a computing system **800** typically includes one or more multicore processor assemblies **801** coupled to volatile memory **802** and a large capacity nonvolatile memory, such as a non-volatile disk drive **804**. The computing system **800** may include network access ports **803** coupled to the multicore processor assemblies **801** for establishing network interface connections with a network **805**, such as a local area network, a vehicle network bus, and other networks.

[0083] Implementation examples are described in the following paragraphs. While some of the following implementation examples are described in terms of example systems, devices, or methods, further example implementations may include: the example systems or devices discussed in the following paragraphs implemented as a method executing operations of the example systems or devices; the example systems, devices, or methods discussed in the following paragraphs implemented by a computing device comprising a processing device and/or a memory subsystem scan dump device configured with processing device-executable instructions to perform operations of the example systems, devices, or methods; the example systems, devices, or methods discussed in the following paragraphs implemented by a memory subsystem scan dump device configured to perform operations of the example systems, devices, or methods; the example systems, devices, or methods discussed in the following paragraphs implemented by a computing device comprising a memory subsystem scan dump device configured to perform operations of the example systems, devices, or methods; the example systems, devices, or methods discussed in the following paragraphs implemented by a computing device including means for performing functions of the example systems, devices, or methods; and the example systems, devices, or methods discussed in the following paragraphs implemented as a non-transitory processor-readable storage medium having stored thereon processor-executable instructions configured to cause a processor of a computing device to perform the operations of the example systems, devices, or methods.

[0084] Example 1. Example 1. A method of implementing command transactions on a two-wire bus system of a computing device, including: receiving, by a two-wire interface of a host device, at least one abort signal via a two-wire bus; and transmitting, by the two-wire interface of the host device, an end signal via the two-wire bus in response to receiving the at least one abort signal.

[0085] Example 2. The method of example 1, further including: transmitting, by a two-wire interface of a bus owner device, at least one command via the two-wire bus prior to receiving the end signal; receiving, by the two-wire interface of the bus owner device, the end signal via the two-wire bus; and terminating, by the two-wire interface of the bus owner device, transmission of commands on the two-wire bus in response to receiving the end signal.

[0086] Example 3. The method of example 2, in which transmitting, by the two-wire interface of the bus owner device, the at least one command via the two-wire bus prior to receiving the end signal includes transmitting a plurality of commands, including the at least one command, following an instance of bus arbitration and prior to a subsequent instance of bus arbitration.

[0087] Example 4. The method of either of examples 2 or 3, in which the bus owner device is a power management integrated circuit (PMIC).

[0088] Example 5. The method of any of examples 1-4, in which receiving, by the two-wire interface of the host device, the at least one abort signal via the two-wire bus includes receiving the at least one abort signal from a two-wire interface of a bus owner device.

[0089] Example 6. The method of any of examples 1-4, in which receiving, by the two-wire interface of the host device, the at least one abort signal via the two-wire bus includes receiving the at least one abort signal from a two-wire interface of at least one non-bus owner device.

[0090] Example 7. The method of any of examples 1-6, in which the at least one abort signal is a voltage transition on at least one wire of the two-wire bus.

[0091] Computer program code or “program code” for execution on a programmable processor for carrying out operations of the various embodiments may be written in a high level programming language such as C, C++, C#, Smalltalk, Java, JavaScript, Visual Basic, a Structured Query Language (e.g., Transact-SQL), Perl, or in various other programming languages. Program code or programs stored on a computer readable storage medium as used in this application may refer to machine language code (such as object code) whose format is understandable by a processor.

[0092] The foregoing method descriptions and the process flow diagrams are provided merely as illustrative examples and are not intended to require or imply that the operations of the various embodiments must be performed in the order presented. As will be appreciated by one of skill in the art the order of operations in the foregoing embodiments may be performed in any order. Words such as “thereafter,” “then,” “next,” etc. are not intended to limit the order of the operations; these words are simply used to guide the reader through the description of the methods. Further, any reference to claim elements in the singular, for example, using the articles “a,” “an” or “the” is not to be construed as limiting the element to the singular.

[0093] The various illustrative logical blocks, modules, circuits, and algorithm operations described in connection with the various embodiments may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and operations have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the claims.

[0094] The hardware used to implement the various illustrative logics, logical blocks, modules, and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose processor, a digital signal processor (DSP), an application-specific integrated circuit (ASIC), a field programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but, in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. Alternatively, some operations or methods may be performed by circuitry that is specific to a given function.

[0095] In one or more embodiments, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored as one or more instructions or code on a non-transitory computer-readable medium or a non-transitory processor-readable medium. The operations of a method or algorithm disclosed herein may be embodied in a processor-executable software module that may reside on a non-transitory computer-readable or processor-readable storage medium. Non-transitory computer-readable or processor-readable storage media may be any storage media that may be accessed by a computer or

a processor. By way of example but not limitation, such non-transitory computer-readable or processor-readable media may include RAM, ROM, EEPROM, FLASH memory, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that may be used to store desired program code in the form of instructions or data structures and that may be accessed by a computer. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of non-transitory computer-readable and processor-readable media. Additionally, the operations of a method or algorithm may reside as one or any combination or set of codes and/or instructions on a non-transitory processor-readable medium and/or computer-readable medium, which may be incorporated into a computer program product.

[0096] The preceding description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the claims. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments and implementations without departing from the scope of the claims. Thus, the present disclosure is not intended to be limited to the embodiments and implementations described herein, but is to be accorded the widest scope consistent with the following claims and the principles and novel features disclosed herein.

Claims

1. A computing device, comprising: a two-wire bus; a host device including a two-wire interface coupled to the two-wire bus; and a bus owner device including a two-wire interface coupled to the two-wire bus, wherein the two-wire interface of the host device is configured to: receive at least one abort signal via the two-wire bus; and transmit an end signal via the two-wire bus in response to receiving the at least one abort signal.
2. The computing device of claim 1, wherein the two-wire interface of the bus owner device is configured to: transmit at least one command via the two-wire bus prior to receiving the end signal; receive the end signal via the two-wire bus; and terminate transmission of commands on the two-wire bus in response to receiving the end signal.
3. The computing device of claim 2, wherein the two-wire interface of the bus owner device is configured to transmit a plurality of commands, including the at least one command, following an instance of bus arbitration and prior to a subsequent instance of bus arbitration.
4. The computing device of claim 2, wherein the bus owner device is a power management integrated circuit (PMIC).
5. The computing device of claim 1, wherein the two-wire interface of the host device is configured to receive the at least one abort signal from the two-wire interface of the bus owner device.
6. The computing device of claim 1, wherein the two-wire interface of the host device is configured to receive the at least one abort signal from a two-wire interface of at least one non-bus owner device.
7. The computing device of claim 1, wherein the at least one abort signal is a voltage transition on at least one wire of the two-wire bus.
8. A method of implementing command transactions on a two-wire bus system of a computing device, comprising: receiving, by a two-wire interface of a host device, at least one abort signal via a two-wire bus; and transmitting, by the two-wire interface of the host device, an end signal via the two-wire bus in response to receiving the at least one abort signal.
9. The method of claim 8, further comprising: transmitting, by a two-wire interface of a bus owner device, at least one command via the two-wire bus prior to receiving the end signal; receiving, by the two-wire interface of the bus owner device, the end signal via the two-wire bus; and

terminating, by the two-wire interface of the bus owner device, transmission of commands on the two-wire bus in response to receiving the end signal.

10. The method of claim 9, wherein transmitting, by the two-wire interface of the bus owner device, the at least one command via the two-wire bus prior to receiving the end signal comprises transmitting a plurality of commands, including the at least one command, following an instance of bus arbitration and prior to a subsequent instance of bus arbitration.

11. The method of claim 9, wherein the bus owner device is a power management integrated circuit (PMIC).

12. The method of claim 8, wherein receiving, by the two-wire interface of the host device, the at least one abort signal via the two-wire bus comprises receiving the at least one abort signal from a two-wire interface of a bus owner device.

13. The method of claim 8, wherein receiving, by the two-wire interface of the host device, the at least one abort signal via the two-wire bus comprises receiving the at least one abort signal from a two-wire interface of at least one non-bus owner device.

14. The method of claim 8, wherein the at least one abort signal is a voltage transition on at least one wire of the two-wire bus.

15. A computing device, comprising: a host device; a bus owner device; a two-wire bus communicatively coupling the host device to the bus owner device; means for receiving by the host device, at least one abort signal via the two-wire bus; and means for transmitting, by the two-wire interface of the host device, an end signal via the two-wire bus in response to receiving the at least one abort signal.

16. The computing device of claim 15, further comprising: means for transmitting by the bus owner device at least one command via the two-wire bus prior to receiving the end signal; means for receiving by the bus owner device the end signal via the two-wire bus; and means for terminating by the bus owner device transmission of commands on the two-wire bus in response to receiving the end signal.

17. The computing device of claim 16, wherein means for transmitting by the bus owner device the at least one command via the two-wire bus prior to receiving the end signal comprises means for transmitting a plurality of commands, including the at least one command, following an instance of bus arbitration and prior to a subsequent instance of bus arbitration.

18. The computing device of claim 16, wherein the bus owner device is a power management integrated circuit (PMIC).

19. The computing device of claim 15, wherein means for receiving by the host device the at least one abort signal via the two-wire bus comprises means for receiving the at least one abort signal from a two-wire interface of the bus owner device.

20. The computing device of claim 15, wherein means for receiving by the host device the at least one abort signal via the two-wire bus comprises means for receiving the at least one abort signal from a two-wire interface of at least one non-bus owner device.
